

Progesterone and Negative Emotionality Across and Between Ovulatory Cycles: A Study of Romantically Involved Women

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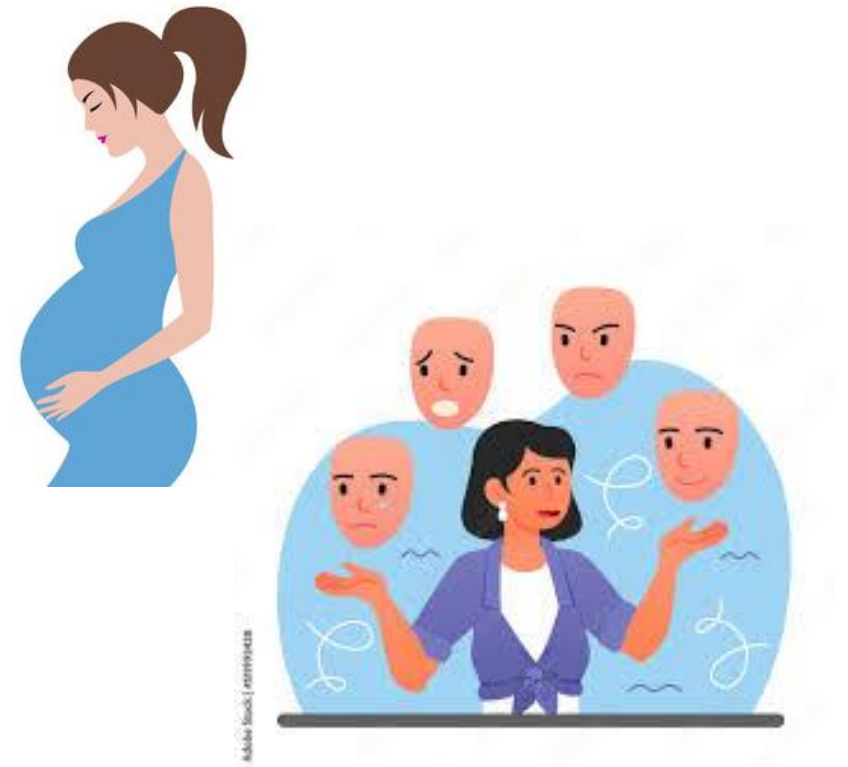
UNIVERSITY OF NEW MEXICO 2ND YEAR PHD STUDENT, EVOLUTION & SOCIETY AREA



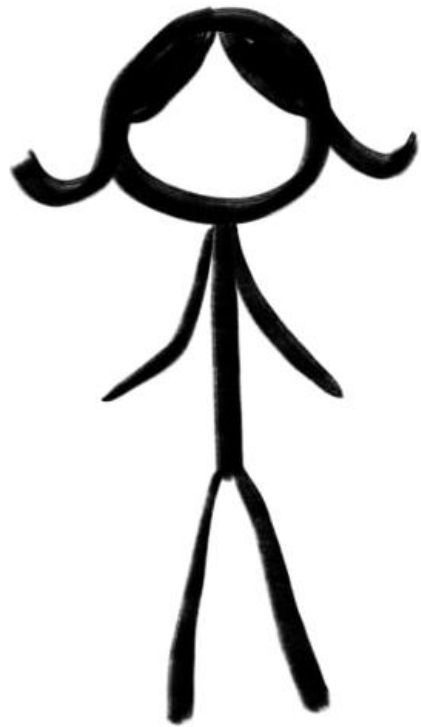
Ovulatory Adaptations

Progesterone and Estrogen orchestrate physiological changes across the menstrual cycle.

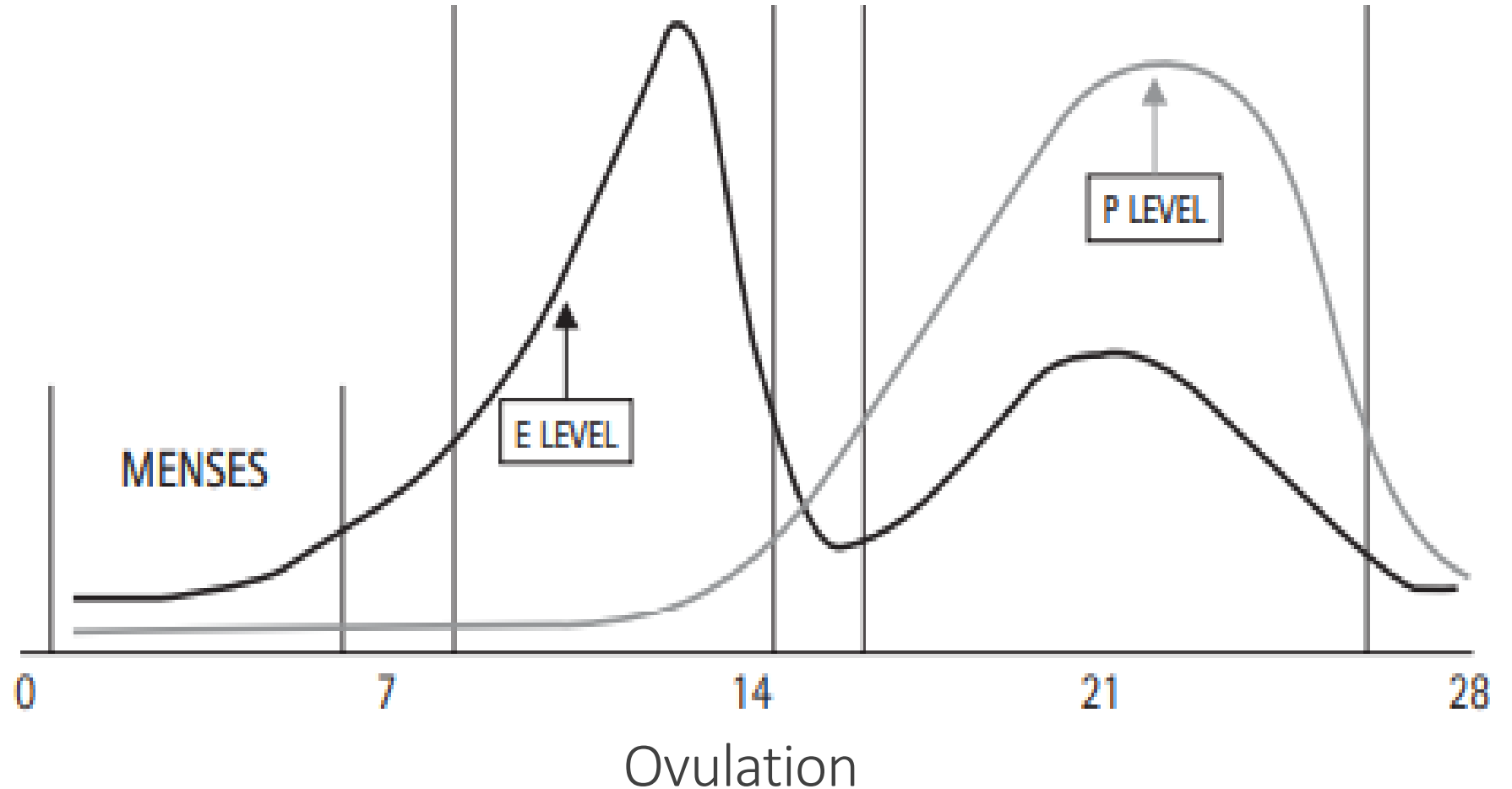
Plausible that the same hormones facilitate psychological changes.



Social Support Vigilance Hypothesis: women post-ovulation should be more concerned about their levels of social support, which is driven by increasing progesterone levels



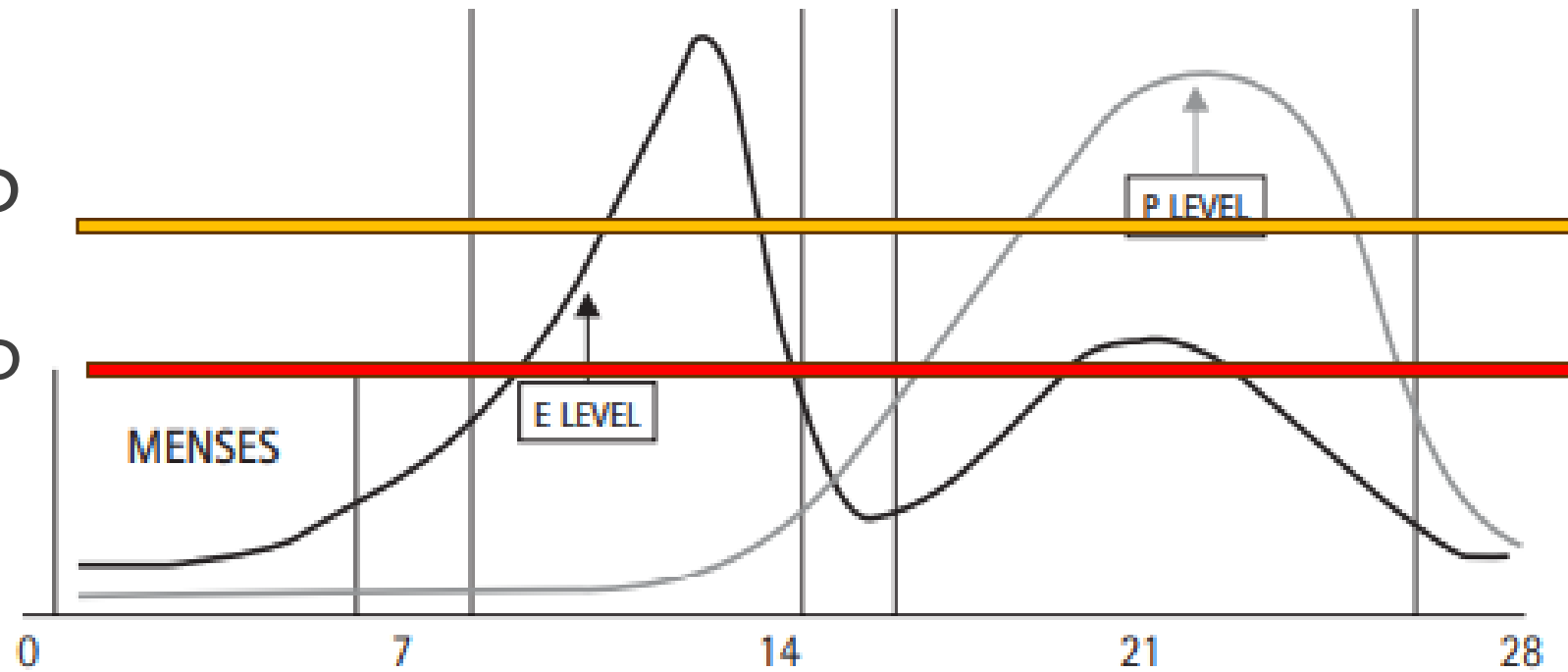
Eve



Between-Women Effects

Sam's Average P

Eve's Average P



The Current Study

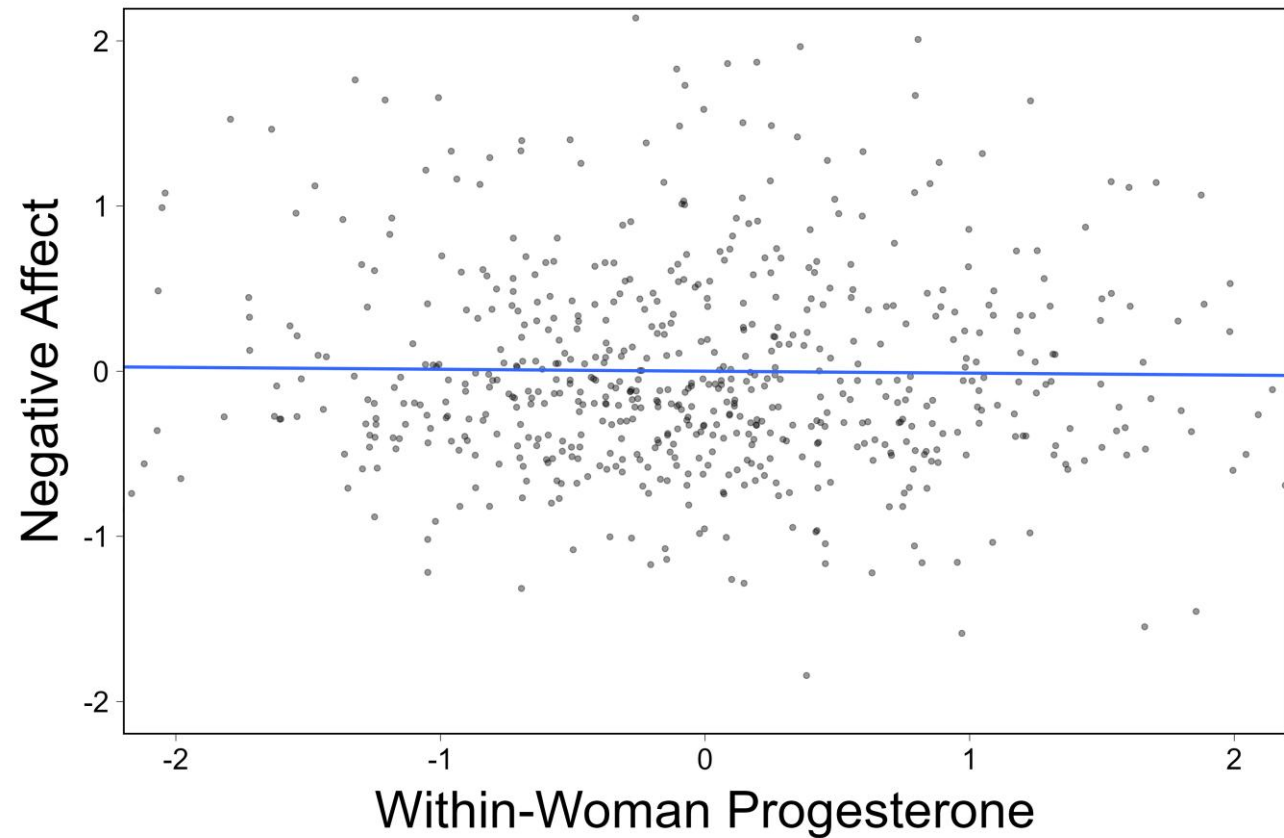
- ❖ 180 women attended between 2 and 4 lab sessions ($M = 3.6$).
- ❖ At each session...
 - ❖ Urine sample to assay hormones: progesterone, estrogen (estradiol), cortisol, and testosterone
 - ❖ Negative Emotionality measure (6 items; e.g., “How much have you felt sad in the last two days”).
 - ❖ Worries about romantic partner’s involvement (3 items; e.g., “I worried about my partner’s involvement.”)

Inclusion Criteria:

- ❖ In a heterosexual romantic relationship
 - ❖ At least 3 months long
- ❖ Age 18-35 ($M = 20.9$)
- ❖ Naturally cycling

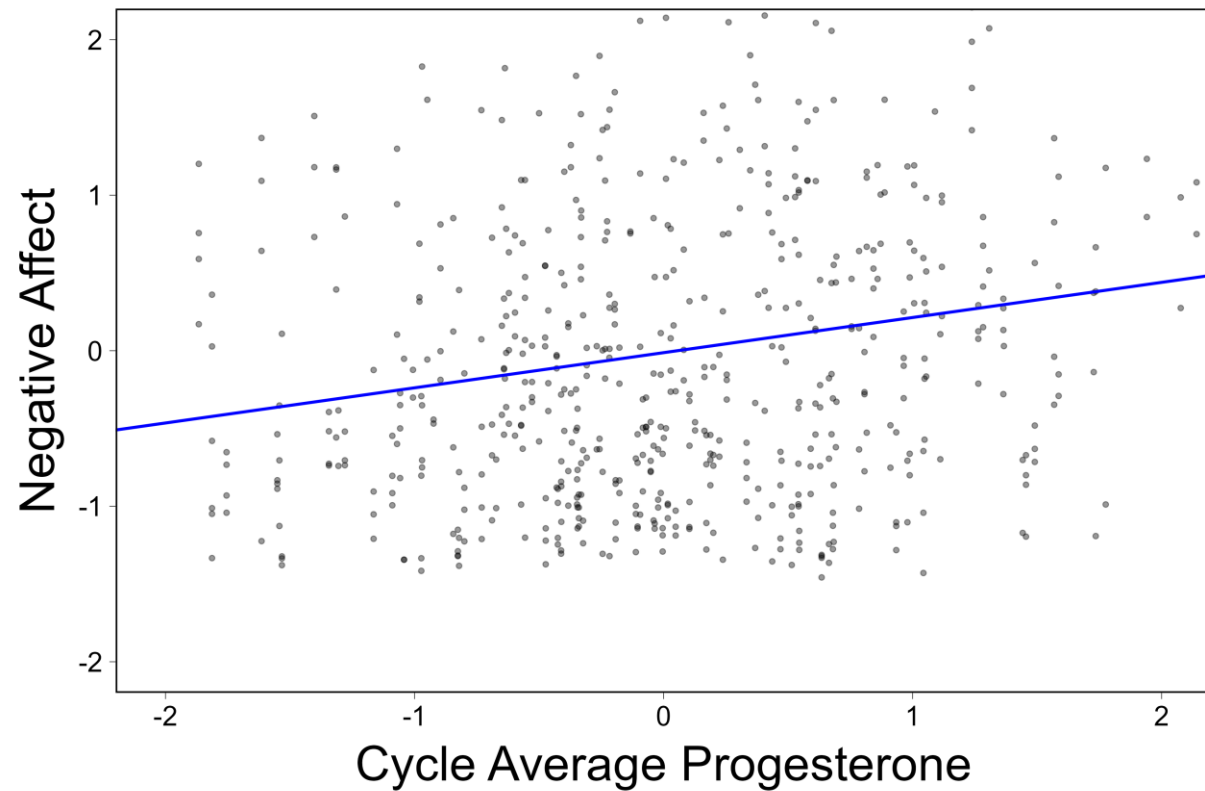
No Within-Women Effect

When “Eve” had higher P levels in her cycle she was not more likely to have more negative affect or worries about her romantic partner’s involvement.



Significant Positive Between-Women Effect

When Sam had a higher cycle average amount of P than Eve, Sam was also more likely to have more negative affect and worries about her romantic partner's involvement.



Confounds?

All results discussed here included controls for:

- Within and between E, C, and T
- Menses

Other analyses included additional controls:

- Weekend effect (no change)
- Age (no change)
- Individual differences (between-women effects 30% smaller)
 - Friend and Family Support
 - Big Five Personality Traits
 - Attachment Styles

What's happening?

- Lack of within-person effect is troubling for social support vigilance hypothesis.
- Unclear what causal process is at play to generate the consistent between-woman association between progesterone and negative affect and worries about romantic partner's relationship involvement.
 - Some additional analyses consistent with reverse causal process for negative affect

Manuscript and Supplemental Materials

